

Article

A Two-Stage Machine Learning Framework for High-Resolution Multi-Source Precipitation Fusion in Complex Terrain: A Case Study of Shaoxing, China

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Abstract

High-resolution precipitation fields are essential for flash-flood forecasting and hydrological risk management, especially in small and medium-sized basins, yet single-source precipitation products often show limited accuracy over complex terrain. This study develops a two-stage machine-learning framework for 1 km/1 h multi-source precipitation fusion over Shaoxing, China, during the 2025 flood season. In the first stage, a machine-learning classifier identifies precipitation occurrence and reduces zero-inflated noise; in the second stage, an optimized tree-based residual-regression model corrects precipitation estimates for rainy samples. A 61-dimensional feature set was constructed by integrating satellite precipitation estimates, weather-radar precipitation estimates from the Zhejiang radar network, temporal-lag and accumulation statistics, neighborhood descriptors, cyclic time variables, and terrain-derived interaction features, with gauge observations used as the training target. After quality control, the dataset comprised 41,458 hourly station samples from 72 rain gauges. The stations were divided at the station level into a 57-station development set and a fixed 15-station held-out spatial test set containing 8637 hourly samples. Station-blocked fivefold cross-validation within the development set was used for model selection, hyperparameter tuning, and probability-threshold selection, whereas the held-out stations were used only for final performance evaluation. On the fixed held-out test set, the occurrence classifier achieved an overall accuracy of 0.947, with a probability of detection of 0.806, a false alarm ratio of 0.158, a critical success index of 0.700, and an F1 score of 0.823. For quantitative estimation, the two-stage fusion product reduced root mean square error from 2.342 mm for satellite precipitation estimates to 1.189 mm, corresponding to a 49.22% reduction, and decreased mean absolute error from 0.712 mm to 0.262 mm, while increasing the coefficient of determination to 0.685. The fused precipitation product also improved the detection of intense rainfall events, with probability of detection and critical success index reaching 0.511 and 0.442, respectively, for events exceeding 10 mm/h, while reducing false weak precipitation and showing closer agreement with observed station-level spatial variability. By separating precipitation-occurrence identification from rainfall-intensity correction, the framework reduces zero-inflated bias, improves heavy-rainfall representation, and demonstrates predictive skill at gauges excluded from model development during the 2025 flood season.



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1. Introduction

Accurate high-resolution precipitation information is a prerequisite for flash-flood warning, hydrological forecasting, and hydrological risk management in small and medium-sized basins. In regions with complex terrain, such as Zhejiang Province in southeastern China, precipitation fields are strongly affected by topographic effects, including orographic enhancement, radar beam blockage, and the sparse distribution of rain gauges in mountainous areas [1–4]. These characteristics make it difficult for single-source precipitation products to accurately represent rainfall variability at the spatial and temporal scales required for flash-flood applications.

Rain gauges provide reliable point-scale observations but cannot fully resolve spatial heterogeneity in mountainous areas because of their sparse spatial coverage. Satellite precipitation products, such as Global Precipitation Measurement (GPM) Integrated Multi-satellitE Retrievals for GPM (IMERG), provide continuous spatial coverage, but they often suffer from smoothing biases, retrieval uncertainties, and underestimation of local extremes at their native resolution [5–14]. Weather radar quantitative precipitation estimation (QPE) offers high spatiotemporal resolution and detailed rainfall structure, but its accuracy can be degraded by beam blockage, ground clutter, and range-dependent attenuation in rugged terrain [15–19]. These limitations highlight the need for precipitation fusion methods that can combine the complementary strengths of rain gauges, satellite estimates, radar QPE, and terrain information.

Multi-source precipitation fusion has become an important strategy for improving rainfall estimation. Early approaches, including optimal interpolation and Bayesian model averaging, established useful statistical frameworks for integrating multiple precipitation sources [20–24], but their assumptions of linearity and stationarity can be difficult to satisfy under complex-terrain and extreme-rainfall conditions [25]. Machine-learning methods, including random forests, gradient-boosting models, and neural networks, have therefore been increasingly adopted to represent nonlinear and high-dimensional relationships between multi-source inputs and gauge observations [26–29]. Two-stage architectures further address the strong zero inflation of precipitation data by separating occurrence detection from intensity estimation [30,31]. Recent advances in complex-terrain precipitation estimation have developed along several complementary directions. Terrain-informed satellite downscaling uses topographic and environmental predictors to estimate finer-scale rainfall variability from coarse satellite products [32]. Radar-centered approaches combine three-dimensional reflectivity, neural networks, interpolation, and gauge correction to improve QPE in mountainous regions [33,34], whereas spatiotemporal tree-based models strengthen satellite–gauge merging by representing temporal persistence and spatial dependence [35]. Together, these studies highlight the utility of terrain, radar, and spatiotemporal information, although these components are often examined within different data combinations, scales, or estimation tasks. Building on these complementary lines of work, the present study integrates satellite precipitation, radar QPE, gauge observations, terrain descriptors, temporal-memory variables, and neighborhood statistics within a 1 km/1 h two-stage framework that separates precipitation-occurrence detection from conditional intensity correction.

Despite these advances, several challenges remain for high-resolution precipitation fusion in complex terrain. First, existing two-stage applications have mainly focused on

coarser temporal scales or different hydroclimatic settings. For example, Senocak et al. [30] targeted numerical weather prediction (NWP)-based precipitation forecasts at a daily scale, whereas Wang et al. [31] focused on monthly drought monitoring in China's drylands using gauge-satellite fusion. It therefore remains unclear whether two-stage architectures can be effectively generalized to kilometer-scale, hourly rainfall estimation for highly localized, flash-flood-producing storms in mountainous regions. Second, complex terrain imposes strong observational constraints on precipitation products, including radar beam blockage, ground clutter, terrain-induced enhancement, and spatially varying retrieval errors. However, these effects are not always explicitly represented in machine learning-based fusion frameworks through terrain-aware feature engineering. Consequently, in complex-terrain settings, fusion products may still underestimate intense rainfall on windward slopes or generate false weak precipitation signals in radar-blind zones. Third, many fusion studies rely on random sample splitting for model evaluation, which may overestimate performance by allowing samples from the same stations to appear in both training and testing datasets. Station-blocked spatial cross-validation is therefore needed to assess model transferability to unseen locations.

To address these gaps, this study develops a terrain-aware two-stage machine learning framework for 1 km/1 h multi-source precipitation fusion over Shaoxing City, Zhejiang Province, China. The proposed framework first uses a based Light Gradient Boosting Machine (LightGBM) classifier to distinguish rainy and non-rainy samples and then applies an eXtreme Gradient Boosting (XGBoost)-based residual regression model to correct precipitation intensity for rainy samples. Multi-source dynamic precipitation inputs are integrated with digital elevation model (DEM)-based static terrain constraints, temporal-lag and accumulation descriptors, spatial-neighborhood statistics, cyclic time variables, and physically interpretable interaction terms to improve the representation of rainfall variability in complex terrain. The specific objectives are: (i) to construct a 1 km/1 h multi-source precipitation fusion framework that explicitly separates precipitation occurrence detection from intensity correction; (ii) to evaluate the added value of terrain-aware and multi-scale feature engineering in complex terrain; (iii) to assess station-level spatial interpolation capability at previously unseen stations within the same flood-season event period using station-blocked validation; and (iv) to explore the potential of the fused precipitation product for identifying gauge-sparse areas and supporting rain gauge network optimization.

2. Study Area and Data

2.1. Study Area

Shaoxing City, with an area of about 8279 km², is located in the transition zone between the southeastern coast of China and the hilly terrain of eastern Zhejiang Province. The city exhibits a pronounced north-south topographic gradient. The southern part is dominated by low-mountain and hilly terrain, with elevations generally ranging from 200 to 1200 m and local slope gradients frequently exceeding 25°, whereas the northern part is characterized by a flat river-network plain with elevations mostly below 10 m [36]. Major river systems include the Puyang River, the Cao'e River, and the lower reaches of the Yongjiang River, which ultimately drain into Hangzhou Bay. The short flow paths, steep hillslopes, and dense river networks make local catchments highly sensitive to short-duration rainfall forcing.

Shaoxing has a subtropical monsoon climate with a distinct flood season from May to September, including the Meiyu period, typically from June to mid-July, and the typhoon season from July to September. Under the combined influence of monsoonal moisture transport, orographic lifting on windward slopes, valley convergence, and funneling effects, localized short-duration heavy rainfall can exceed 50 mm/h in mountainous areas [1,36].

These rainfall characteristics pose considerable challenges for flash-flood forecasting and hydrological risk management at hourly and kilometer scales.

The 72 rain gauges used in this study are unevenly distributed across the study area. Stations are relatively concentrated in the northern plain and along river valleys, with a station density of approximately one gauge per 50 km² in the northern lowland region, but are much sparser in the southern mountainous areas, where the density is generally lower than one gauge per 200 km². This uneven observational network is typical of complex-terrain regions and implies that precipitation errors from satellite and radar products in the southern mountains cannot be fully constrained by gauge observations alone. Therefore, Shaoxing provides a suitable testbed for evaluating terrain-aware multi-source precipitation fusion methods.

The suitability of Shaoxing for this study is mainly reflected in two aspects. First, precipitation errors from GPM IMERG, radar QPE, and gauge-based information are spatially heterogeneous and physically related to topography, radar coverage, and rain-gauge distribution. Second, the coexistence of mountainous data-sparse areas and relatively well-observed plains allows the station-level spatial interpolation capability of the fusion framework to be evaluated under realistic observational constraints. The topography of Shaoxing, the spatial distribution of rain gauges, and the radar QPE coverage are shown in Figure 1.

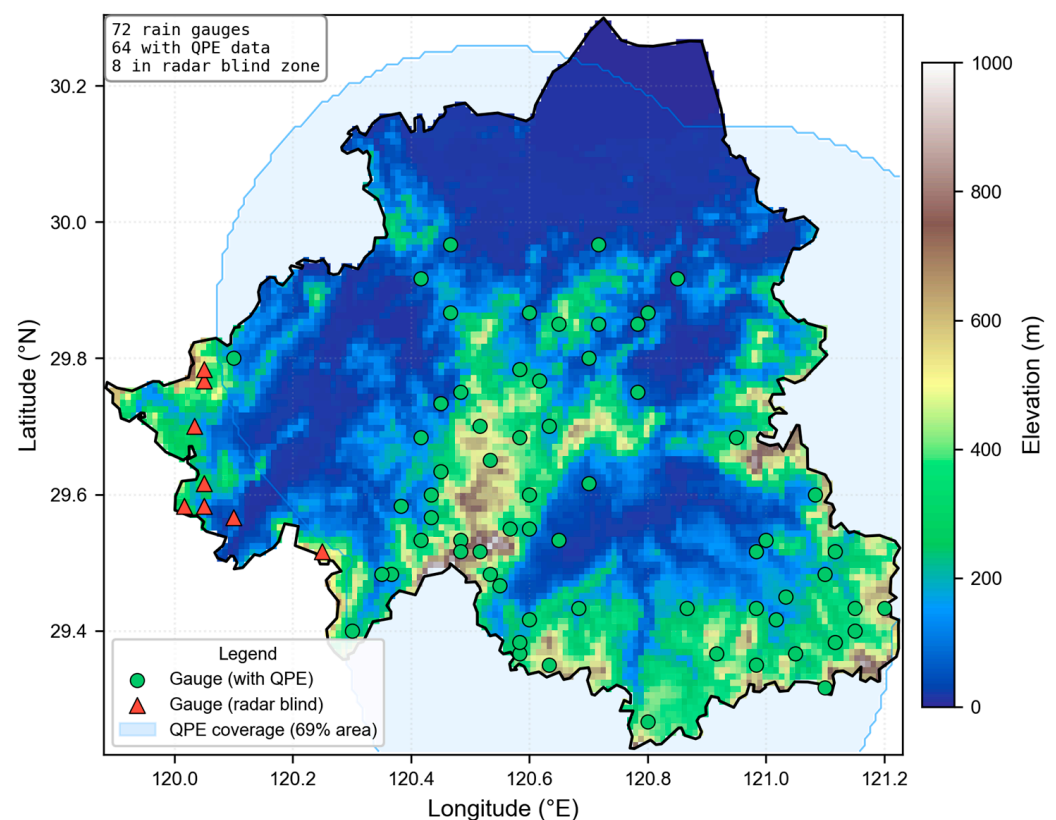


Figure 1. Topography of Shaoxing, spatial distribution of 72 rain gauges, and radar QPE coverage footprint. Green circles indicate gauges with valid QPE coverage, red triangles indicate gauges located in radar-blind zones, and the blue outline indicates the radar QPE coverage area. The QPE coverage percentage refers to areal radar coverage, whereas the QPE availability rate reported in the text refers to valid station-hour samples.

2.2. Data Sources and Preprocessing

Four categories of data were used to construct the multi-source precipitation fusion dataset, including rain-gauge observations, satellite precipitation, radar quantitative precipitation estimation (QPE), and digital elevation data. For each data source, quality control, temporal aggregation, and spatial matching were performed according to its original resolution and data characteristics. After preprocessing, all variables were aligned to a common hourly time step and a unified 1 km target grid.

Rain-gauge observations were obtained from 72 national and regional automatic meteorological stations within Shaoxing. The original hourly precipitation records were subjected to quality-control procedures, including extreme-value checks, internal consistency checks, spatial consistency control, and missing-value screening, following the procedures commonly used for automatic weather station precipitation data [37]. Hourly records containing missing values, as well as negative or physically unrealistic precipitation values, were regarded as invalid and excluded rather than infilled, thereby ensuring that only reliable observations were used for model development. The quality-controlled gauge observations were used as reference targets for model training and validation. A threshold of 0.1 mm h^{-1} was used to distinguish rainy and non-rainy samples in the precipitation occurrence classification task.

Satellite precipitation was represented by the GPM IMERG Late Run V07 product (3IMERGHHL). The original half-hourly precipitation rate fields at 0.1° spatial resolution were accumulated to hourly totals and then bilinearly interpolated to the 1 km target grid. The interpolated IMERG fields were used as large-scale precipitation background predictors rather than independent 1 km observations, because spatial interpolation does not create new fine-scale rainfall structures. This treatment preserves regional rainfall information while avoiding overinterpretation of the downscaled satellite field. Considering that IMERG can be affected by retrieval smoothing and coarse-resolution biases in complex terrain, it was further corrected within the multi-source fusion framework [6,38].

Radar QPE was generated from the Zhejiang S-band Doppler radar network, which consists of three operational radars covering Shaoxing and its surrounding areas [16,18,19]. The native QPE fields are produced at 10 min intervals, with the raw variable TenMP representing precipitation accumulation in millimeters per 10 min. For each clock hour, all available 10 min files were summed to obtain hourly accumulations. When four or five of the six expected files were available, the hourly accumulation was proportionally adjusted by a factor of $6/N_{i,t}$, where N is the number of valid 10 min files. The hourly radar accumulation was calculated as

$$R_{i,t}^{1h} = \frac{6}{N_{i,t}} \sum_{j=1}^{N_{i,t}} R_{i,t,j}^{10 \text{ min}}, N_{i,t} \geq 4 \quad (1)$$

where $N_{i,t}$ denotes the number of valid 10 min radar files within hour t . Hours with fewer than four valid files were treated as missing to avoid introducing large uncertainty during intense rainfall periods. Missing or invalid raw values, originally encoded as -1.0 , were converted to NaN and excluded from the hourly accumulation rather than interpolated. The hourly radar fields were then resampled to the unified 1 km grid while preserving their fine-scale spatial structure. A binary availability flag, QPE_available, was assigned to each grid cell and hour: a value of 1 indicates a valid radar retrieval, including true zero precipitation, whereas 0 indicates no radar coverage, failed retrieval, or insufficient hourly completeness. Across all station-hour samples, the QPE availability rate was 88.9%. For compatibility with the tree-based models, unavailable QPE values were numerically filled with 0 mm only after preprocessing, while the QPE_available flag was retained as an explicit predictor. This strategy distinguishes unavailable radar information from

genuine no-rain conditions without introducing artificial precipitation estimates. Radar QPE provides fine spatial detail and strong sensitivity to convective rainfall cores, but its accuracy may decline in radar-shadow zones and areas affected by beam blockage and ground clutter, particularly in the southern mountainous part of the study area [39].

Terrain information was derived from the 90 m Shuttle Radar Topography Mission (SRTM) digital elevation model (V4.1). Elevation and slope were extracted and aggregated to the 1 km target grid. These variables were used as first-order static terrain descriptors and as the basis for terrain-interaction features in the subsequent feature engineering step. Such DEM-derived information helps represent the influence of local topography on precipitation enhancement, observation representativeness, and terrain-dependent retrieval errors [40].

Finally, gridded satellite and radar predictors were matched to the gauge records at the station coordinates. For each gauge-hour, the temporally corresponding IMERG and radar QPE fields were selected using the nearest hourly timestamp within a 30 min tolerance, and the precipitation values at the gauge longitude and latitude were extracted using bilinear interpolation from the 1 km grids. Gauge observations were retained as point-scale reference targets, whereas DEM-derived terrain variables were extracted from the corresponding terrain grid. After quality control, temporal aggregation, and spatial resampling, a standardized multidimensional data cube containing gauge, satellite, radar, and terrain variables was produced for subsequent feature extraction and model development. Remaining NaN values in derived predictors occurred primarily at the beginning of each station record because insufficient historical observations were available to compute lagged variables or moving-window statistics. The corresponding samples were retained in the dataset, and the missing derived values were filled with 0 before model training for compatibility with the tree-based algorithms. The only explicit availability indicator included in the feature matrix was QPE_available, which distinguishes unavailable radar QPE from genuine zero precipitation.

3. Method

The proposed framework was designed to address the zero-inflated and highly skewed distribution of hourly precipitation. Instead of directly fitting a single regression model to all samples, precipitation fusion was decomposed into two sequential tasks: occurrence classification and conditional intensity correction. In the first stage, precipitation occurrence was treated as a binary classification problem. A gauge observation was labelled as rainy when the recorded precipitation was equal to or greater than 0.1 mm h^{-1} ; otherwise, it was labelled as non-rainy. This threshold was used only to construct the observed occurrence label. The LightGBM classifier then produced a continuous occurrence probability, and a separate probability cutoff was optimized on validation data to convert this probability into a rain/no-rain prediction.

In the second stage, precipitation intensity was corrected only for samples classified as rainy by the first-stage model. The final fused precipitation was set to zero for predicted non-rainy samples and was estimated by the XGBoost intensity-correction model for predicted rainy samples. This two-stage design reduces the dominance of zero values in the regression loss, suppresses false weak precipitation caused by satellite retrieval noise or radar clutter, and allows the intensity model to focus on reconstructing the nonlinear amplitude of true rainfall events. The overall workflow of the proposed two-stage precipitation fusion framework is illustrated in Figure 2.

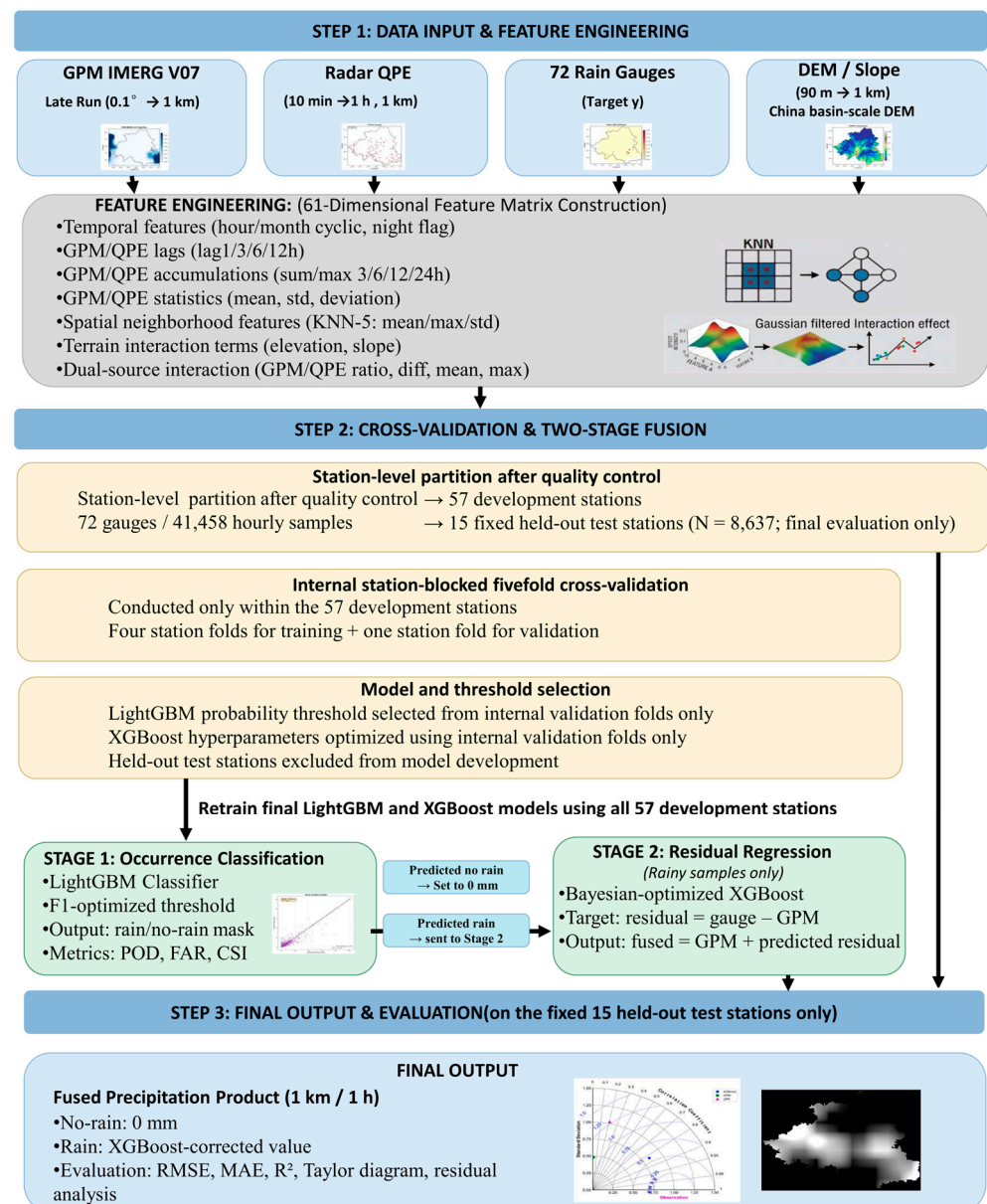


Figure 2. Overall two-stage fusion and validation workflow. The 72 rain gauges were first split into 57 development stations and 15 fixed held-out test stations. Station-blocked fivefold cross-validation, threshold selection, and hyperparameter tuning were conducted only within the development set, followed by final retraining on all development stations and one final evaluation on the held-out test set.

3.1. Terrain-Aware Multi-Scale Feature Engineering

Terrain can substantially modulate rainfall in mountainous and hilly regions [41]. Moist low-level airflow forced to ascend over mountains undergoes adiabatic cooling and condensation, which can enhance precipitation on windward slopes. Elevation modifies the vertical structure of moisture and temperature, while slope affects the strength of forced ascent and the spatial organization of rainfall. To enable the model to represent these effects, a 61-dimensional feature matrix was constructed by integrating static terrain descriptors, dynamic multi-source precipitation states, temporal-memory variables, neighborhood statistics, and physically interpretable interaction terms.

All temporal predictors were calculated using information available at the current hour or previous hours only, so that no future information was introduced into model training or evaluation. Lagged variables were calculated at 1, 3, 6 and 12 h before the target

hour. Moving-window accumulation, maximum, mean and standard deviation statistics were calculated over 3, 6, 12 and 24 h windows ending at the target hour. Neighborhood statistics were derived from the surrounding 1 km grid cells using K-nearest-neighbor statistics. Missing values caused by insufficient temporal history at the beginning of records were filled with zero; this treatment was applied to lagged variables, moving-window statistics, and neighborhood features.

The feature matrix was organized into static geographic features, cyclic time features, multi-source instantaneous fields, temporal-lag descriptors, accumulation and extreme-value statistics, local-neighborhood spatial statistics and multi-source/terrain interaction variables. This design allows the model to represent the rainfall field from point, neighborhood and gridded perspectives. A complete list of the 61 predictors, including their feature groups, symbols, units, definitions and calculation windows, is provided in Appendix A (Table A1).

For each precipitation source x (GPM IMERG or radar QPE), lagged predictors and moving-window statistics were calculated using only information available at or before the target hour:

$$x_{i,t-k}, \quad k = 1, 3, 6, 12 \quad (2)$$

$$A_{i,t}^{(\Delta)} = \sum_{j=0}^{\Delta-1} x_{i,t-j}, \quad \Delta = 3, 6, 12, 24 \quad (3)$$

$$M_{i,t}^{(\Delta)} = \max_{0 \leq j < \Delta} x_{i,t-j}, \quad \Delta = 3, 6, 12, 24 \quad (4)$$

$$\mu_{i,t}^{(24)} = \frac{1}{24} \sum_{j=0}^{23} x_{i,t-j}, \quad \sigma_{i,t}^{(24)} = \sqrt{\frac{1}{24} \sum_{j=0}^{23} (x_{i,t-j} - \mu_{i,t}^{(24)})^2} \quad (5)$$

Spatial-neighborhood descriptors were calculated from the five nearest 1 km grid cells surrounding station i :

$$\bar{x}_{i,t}^{(5)} = \frac{1}{|N_5(i)|} \sum_{l \in N_5(i)} x_{l,t}, \quad M_{i,t}^{(5)} = \max_{l \in N_5(i)} x_{l,t} \quad (6)$$

Terrain-interaction and dual-source predictors were constructed as deterministic transformations of source precipitation, normalized elevation z_i , and normalized slope s_i :

$$D_{i,t}^{QG} = P_{i,t}^{QPE} - P_{i,t}^{GPM}, \quad R_{i,t}^{QG} = \frac{P_{i,t}^{QPE}}{P_{i,t}^{GPM} + \epsilon} \quad (7)$$

$$P_{i,t}^{mean} = \frac{P_{i,t}^{QPE} + P_{i,t}^{GPM}}{2}, \quad P_{i,t}^{max} = \max(P_{i,t}^{QPE}, P_{i,t}^{GPM}) \quad (8)$$

$$T_{i,t}^{GPM} = P_{i,t}^{GPM} z_i, \quad T_{i,t}^{QPE} = P_{i,t}^{QPE} z_i, \quad S_{i,t}^{QPE} = P_{i,t}^{QPE} s_i \quad (9)$$

Here, i denotes the station or matched 1 km grid cell, t denotes the target hour, Δ denotes the moving-window length in hours, $N_5(i)$ denotes the five nearest neighboring grid cells, and ϵ is a small positive constant used to avoid division by zero in ratio features.

3.2. Stage 1: LightGBM Precipitation Occurrence Classifier

Hourly precipitation samples are strongly imbalanced because non-rainy samples greatly outnumber rainy samples [42,43]. If all samples are directly used in a single regression model, the loss function tends to be dominated by zeros, which can lead to underestimated rainfall peaks and false drizzle. Therefore, the first stage transformed quantitative precipitation estimation into a binary occurrence-classification task [30].

Let x_i denote the 61-dimensional feature vector of sample i and $y_i \in \{0, 1\}$ denote the observed occurrence label, where 0 indicates no rain and 1 indicates rain. LightGBM estimates the occurrence probability $p_i = P(y_i = 1 | x_i)$. The model combines multiple

classification and regression tree (CART) learners and optimizes a binary cross-entropy loss. Its histogram-based split strategy and gradient-based one-sided sampling make it efficient for high-dimensional meteorological feature matrices [44].

Because a fixed probability threshold of 0.5 is not necessarily optimal under class imbalance, the decision threshold τ was selected by maximizing the F1 score on the internal validation folds within the 57-station development set. Precision, recall, and F1 were calculated from the numbers of true positives (TP), false positives (FP), and false negatives (FN), as follows: Precision = TP/(TP + FP), Recall = TP/(TP + FN), and F1 = 2 × Precision × Recall/(Precision + Recall). The selected threshold was then used to generate the rain/no-rain mask for the second-stage regression model [45]. To avoid information leakage, the threshold was not optimized on the fixed 15-station held-out spatial test set.

3.3. Stage 2: XGBoost Intensity Correction and Bias Reduction

After occurrence screening, the second stage performed quantitative correction for samples predicted as rainy. XGBoost was selected as the regression model because it can represent nonlinear interactions among satellite background fields, radar QPE structures, temporal-memory descriptors, and terrain constraints [27,28]. The model uses an additive tree ensemble and combines a prediction loss, here the mean squared error, with regularization terms that penalize tree complexity and reduce overfitting [46].

In this study, XGBoost was used as an intensity-correction model rather than a purely unconditional rainfall regressor. For rainy training samples, the residual target was defined as the difference between gauge-observed hourly precipitation and the corresponding GPM IMERG estimate. In the present two-stage fusion framework, the final corrected intensity was constrained to be non-negative and was assigned only to samples classified as rainy. The second-stage correction was formulated as a conditional residual-regression problem. The observed occurrence label, classifier probability, binary rain/no-rain prediction, residual target, and final fused precipitation estimate were defined as follows:

$$o_i = I(y_i \geq 0.1) \quad (10)$$

$$p_i = f_{LGBM}(x_i), \delta_i = I(p_i \geq \tau) \quad (11)$$

$$r_i = y_i - P_i^{GPM} \quad (12)$$

$$\hat{r}_i = f_{XGB}(x_i) \quad (13)$$

$$\hat{y}_i = \delta_i \max(0, P_i^{GPM} + \hat{r}_i) \quad (14)$$

Thus, predicted non-rainy samples were assigned 0 mm, whereas predicted rainy samples were corrected by adding the XGBoost-predicted residual to the GPM background estimate and truncating negative values to zero. This coupling of classification and regression allows the algorithm to avoid the mean-pulling effect of dry samples while preserving sensitivity to heavy rainfall [47].

Bayesian optimization was applied to tune the XGBoost hyperparameters using a Gaussian-process surrogate model and an expected-improvement acquisition function [48,49]. Six key hyperparameters were optimized simultaneously: max_depth $\in$ [4, 12], learning_rate $\in$ [0.01, 0.3], n_estimators $\in$ [100, 600], subsample $\in$ [0.6, 1.0], colsample_bytree $\in$ [0.6, 1.0], and min_child_weight $\in$ [1, 10]. The optimization objective was to minimize the mean absolute error (MAE) of rainy samples on the internal validation folds within the 57-station development set, rather than on the fixed held-out spatial test set. The optimization procedure started with eight initial random evaluations, followed by 20 guided iterations, giving 28 model evaluations in total. The selected configuration was then used to train the final XGBoost regressor on all rainy samples

from the 57 development stations. All model training and spatial preprocessing were implemented in Python 3.13.3 using LightGBM 4.6.0, XGBoost 3.0.2, scikit-learn 1.7.0, bayesian-optimization 3.1.0, rasterio 1.4.3, GeoPandas 1.1.1, and GDAL 3.10.3.

The search ranges were selected to balance model flexibility, overfitting control, and computational feasibility for the size and imbalance of the rainy-sample training set. The range of max_depth from 4 to 12 allowed the model to represent nonlinear interactions among satellite, radar, temporal, neighborhood, and terrain predictors while avoiding excessively deep trees that could memorize station-specific noise. The learning_rate range of 0.01–0.3 covered conservative boosting updates as well as faster convergence settings commonly used in gradient-boosted tree models, whereas n_estimators from 100 to 600 provided sufficient boosting iterations without making the Bayesian search computationally prohibitive. The subsample and colsample_bytree ranges of 0.6–1.0 were used to introduce stochastic regularization and reduce dependence on any single subset of rainy samples or predictors. Finally, min_child_weight was searched from 1 to 10 to control the minimum amount of information required for a leaf split, thereby limiting overly specific corrections for rare heavy-rainfall samples. These ranges were therefore intended to cover plausible model-complexity and regularization settings while keeping all tuning decisions within the internal validation folds of the 57-station development set.

3.4. Evaluation Metrics and Validation Strategy

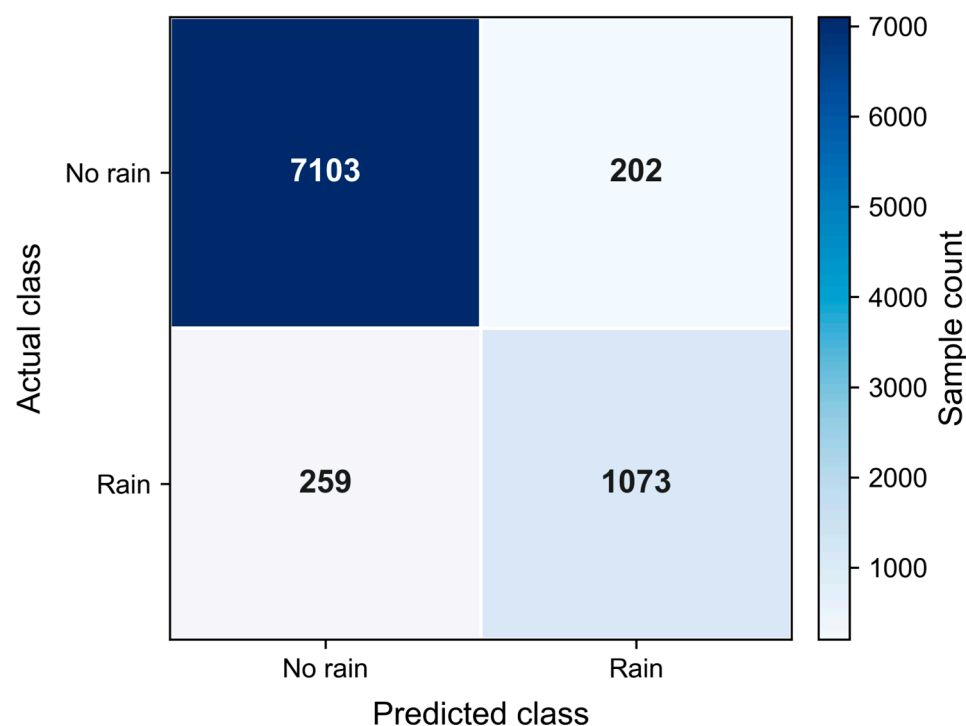
Model performance was assessed from two perspectives: precipitation occurrence detection and quantitative precipitation estimation. For occurrence classification, probability of detection (POD), false alarm ratio (FAR), and critical success index (CSI) were used. Given hits H , misses M , and false alarms F , these metrics are defined as $POD = H/(H + M)$, $FAR = F/(H + F)$, and $CSI = H/(H + M + F)$ [50].

For quantitative estimation, the correlation coefficient (CC), root mean square error (RMSE), mean absolute error (MAE), coefficient of determination (R^2), and bias were calculated between fused precipitation and gauge observations. RMSE emphasizes large errors and is particularly relevant for evaluating heavy-rainfall correction, whereas MAE provides a more robust measure of average absolute error. The quantitative metrics were calculated for all samples, rainy samples, and heavy-rainfall subsets to evaluate both general performance and the ability to reconstruct intense precipitation. Heavy-rainfall samples can be defined using a fixed threshold or an upper-percentile threshold such as the 95th or 99th percentile of gauge-observed hourly precipitation.

After quality control, the dataset contained 41,458 hourly station samples from 72 rain gauges. The stations were partitioned by station location, rather than by individual hourly records, into a 57-station development set and a fixed 15-station held-out spatial test set containing 8637 hourly samples. Within the development set, station-blocked fivefold cross-validation was used for model development and selection [51,52]. In each internal cycle, four station folds were used for training, and the remaining station fold was used for validation, with all records from a given gauge retained in the same fold. XGBoost hyperparameter tuning and LightGBM probability-threshold selection were performed exclusively using these internal validation folds. After the model configuration and threshold had been fixed, the final LightGBM classifier and XGBoost regressor were retrained using all 57 development stations and evaluated once on the 15 held-out stations. Final performance was quantified on this fixed held-out test set, and the corresponding results are reported in Section 4, Table 1, and Figure 3.

Table 1. Quantitative performance of single-source and fused precipitation products.

Product	RMSE (mm)	MAE (mm)	R ²	Main Interpretation
GPM IMERG	2.342	0.712	−0.220	Coarse-resolution smoothing and weak-rain false signals
Radar QPE	2.337	0.565	−0.289	Terrain blockage and local underestimation in mountainous areas
Two-stage XGBoost fusion	1.189	0.262	0.685	Improved peak capture and reduced systematic bias

**Figure 3.** Confusion matrix of the precipitation-occurrence classifier on the fixed 15-station held-out spatial test set (N = 8637 hourly samples). Model configuration and probability-threshold selection were completed within the 57-station development set before final evaluation.

Because the development and held-out stations sampled the same regional events during the 2025 flood season, the held-out evaluation assesses station-level spatial interpolation at gauges excluded from model development. It does not constitute an independent test across years, storm events, or climatic regions.

4. Results

4.1. Precipitation Occurrence Detection

The first-stage LightGBM classifier was evaluated on the fixed 15-station held-out spatial test set, which contained 8637 hourly samples. Using the decision threshold selected within the 57-station development set, the classifier achieved an overall accuracy of 0.947 and a positive-class F1 score of 0.823, with precision and recall of 0.842 and 0.806, respectively. Figure 3 summarizes these final held-out predictions. These results characterize precipitation-occurrence detection at gauges excluded from model development during the 2025 flood season.

As shown in Figure 4, comparison across rainfall-intensity thresholds further showed that the two-stage framework improved event detection relative to single-source GPM and radar QPE. GPM tended to generate false weak precipitation because of satellite retrieval smoothing, while radar QPE suffered from false alarms and missed detections in complex terrain. The LightGBM classifier served as an adaptive digital mask that suppressed false precipitation signals caused by satellite retrieval noise and radar clutter. For moderate and

heavy rainfall categories, the fused framework maintained higher CSI and POD than the single-source products.

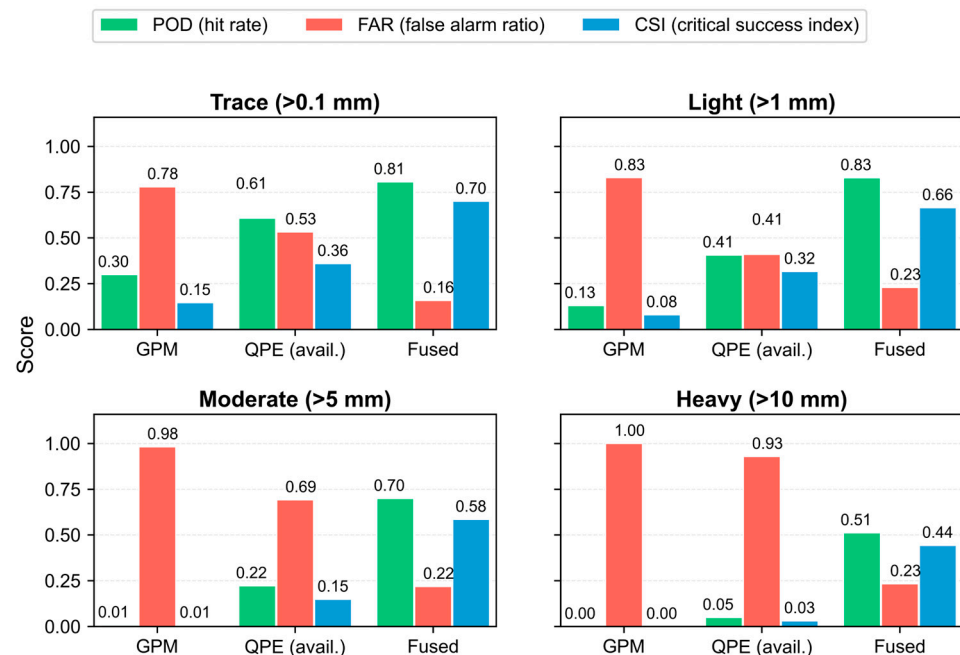


Figure 4. Comparison of precipitation-event detection skill for different rainfall-intensity thresholds.

4.2. Quantitative Precipitation Estimation and Extreme-Value Correction

Results showed a substantial improvement of the proposed two-stage framework over single-source products. Specifically, the original GPM product exhibited an RMSE of 2.342 mm, an MAE of 0.712 mm, and an R^2 of -0.220 . Similarly, the available radar QPE product ($N = 8061$) yielded an RMSE of 2.337 mm, an MAE of 0.565 mm, and an R^2 of -0.289 . In contrast, the proposed two-stage model reduced these errors, achieving an RMSE of 1.189 mm, corresponding to a 49.22% reduction relative to the baseline GPM. The framework also increased R^2 to 0.685, indicating improved representation of nonlinear precipitation residuals within the present Shaoxing test set. The quantitative performance of the single-source products and the fused precipitation product is summarized in Table 1. The GPM IMERG and radar QPE values used in this comparison were the station-matched values extracted with the same bilinear interpolation procedure as used for model input construction. Therefore, Table 1 compares gauge observations with station-level IMERG estimates, station-level radar QPE estimates, and the final two-stage fusion output at the same gauge-hour locations.

The interpolation step introduces an additional spatial-matching uncertainty because a point rain-gauge observation is compared with precipitation fields represented on a 1 km grid. This uncertainty cannot be completely separated from retrieval error, representativeness error, temporal aggregation error, and model error using the present dataset. As a sensitivity check, bilinear station extraction was compared with nearest-neighbor extraction on the held-out test set. The bilinear-minus-nearest difference was very small for GPM, with an RMSE-scale difference of approximately 0.016 mm. For radar QPE, after excluding isolated extreme artifact values removed during quality control, the corresponding RMSE-scale difference was approximately 0.146 mm, and the 95th percentile absolute difference was about 0.058 mm. These values are much smaller than the product-to-gauge errors reported in Table 1, suggesting that interpolation contributes a relatively minor component to the overall uncertainty, whereas retrieval limitations, radar blockage, scale mismatch, and model residual errors dominate the final rainfall-estimation uncertainty.

The scatter-density plots (Figure 5) show a substantial improvement of the fused estimates over the original products across the full intensity range. Overall percent bias (PBIAS) improved from -39.5% (GPM) to -1.4% (fused), indicating that the near-zero mean bias of the fusion product is not merely a cancellation of positive and negative errors.

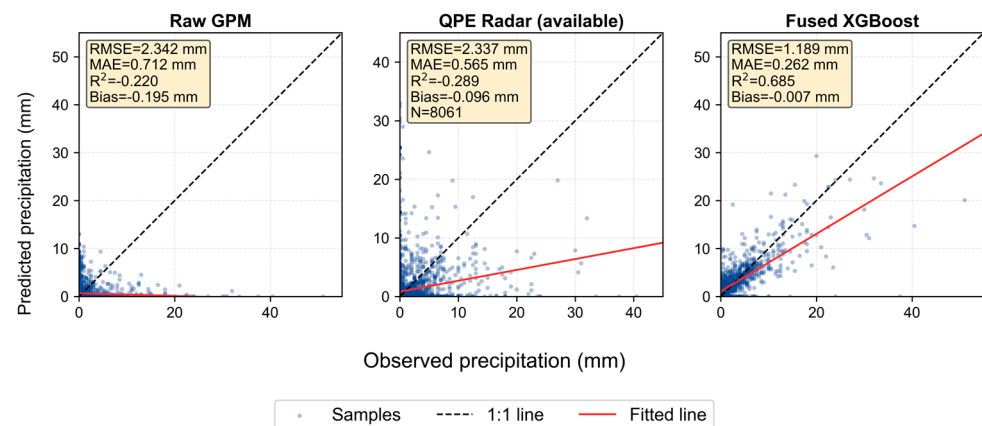


Figure 5. Scatter-density comparison between gauge observations and different precipitation products.

Low-value false alarms were sharply reduced: for samples with no observed precipitation (<0.1 mm), GPM produced false rain for 36.9% of cases ($2697/7305$), whereas the fusion model reduced this to 2.8% ($202/7305$)—a 92% reduction—owing to the first-stage LightGBM classifier.

High-value underestimation was also markedly corrected. Stratified by intensity, the GPM RMSE for light ($1\text{--}5$ mm), moderate ($5\text{--}10$ mm) and heavy (>10 mm) rainfall was 2.87 , 6.89 and 18.61 mm, respectively, whereas the fused RMSE decreased to 1.88 , 2.98 and 9.65 mm. Correspondingly, the mean bias improved from -2.42 to $+0.06$ mm (light), from -6.67 to -1.28 mm (moderate), and from -16.86 to -5.19 mm (heavy). For heavy rainfall, the XGBoost model recovered a mean estimate of 12.15 mm against an observed mean of 17.34 mm, compared to GPM which predicted only 0.48 mm on average.

The residual heavy-rainfall bias of -5.19 mm reflects the remaining challenge of reconstructing extreme convective cores from smoothed satellite fields and terrain-degraded radar signals—a limitation that future work may address by incorporating additional atmospheric variables (Section 5.4).

These results indicate that the two-stage framework reduced average errors and altered the residual distribution in the fixed held-out test set, mainly by suppressing false drizzle at the low end and partially recovering heavy-rainfall magnitude at the high end. This asymmetric correction is consistent with the separation of occurrence detection and intensity regression, which reduces the influence of zero-inflated samples on the regression stage and allows the intensity model to focus on rainy samples.

4.3. Comprehensive Performance Shown by the Taylor Diagram

The Taylor diagram was used to jointly evaluate correlation, centered RMSE and standard deviation ratio. The XGBoost fusion model was located closest to the observational reference point, indicating the most balanced performance in terms of temporal-spatial correlation and variance reproduction. This result suggests that the fused product not only reduced average error but also more closely matched the observed variability of gauge rainfall within the fixed held-out test set. The Taylor diagram in Figure 6 shows that the XGBoost fusion model is located closest to the observational reference point.

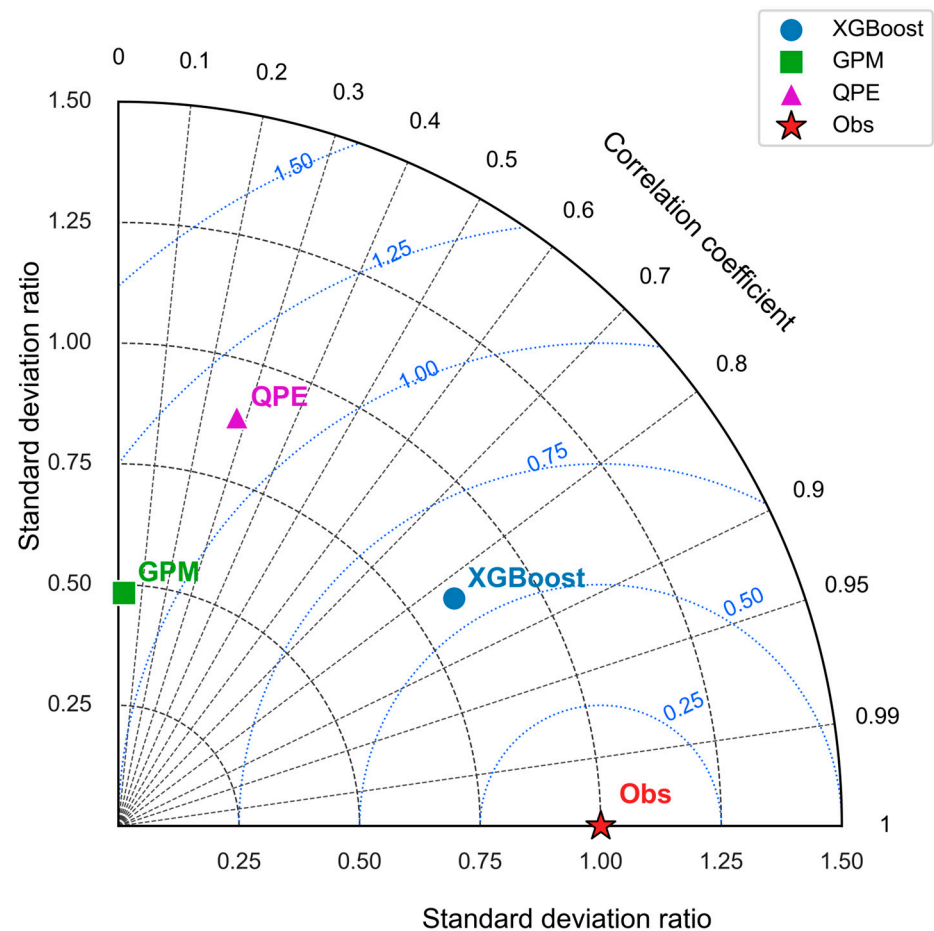


Figure 6. Taylor diagram of GPM, radar QPE and the XGBoost fusion product. The reference point (Obs) marks the observed standard deviation; concentric blue contours show centered root mean square (RMS) difference.

4.4. Residual Distribution and Correction of Heavy Rainfall

Residual analysis revealed clear nonlinear errors in the original single-source products. Weak-rain categories were associated with frequent false alarms, whereas rainfall above 10 mm showed systematic underestimation. After two-stage fusion, residual medians across rainfall grades were close to zero, and the interquartile ranges became much narrower. This indicates that the model achieved nearly unbiased correction across multiple intensity ranges and improved the reconstruction of heavy-rainfall peaks. The residual distributions before and after fusion are shown in Figure 7.

4.5. Event-Scale Time-Series Comparison at an Illustrative Held-Out Station

Figure 8 presents an event-scale illustration at the held-out Station 70154550, including the complete 576 h evaluation period and the corresponding Stage-1 occurrence outcomes. Over the full record, the fused product achieved an RMSE of 0.672 mm and an R^2 of 0.842. At the occurrence-detection stage, 75 of 89 observed wet hours were detected and 471 of 487 observed dry hours were correctly classified, yielding POD, FAR, and CSI values of 0.843, 0.176, and 0.714, respectively.

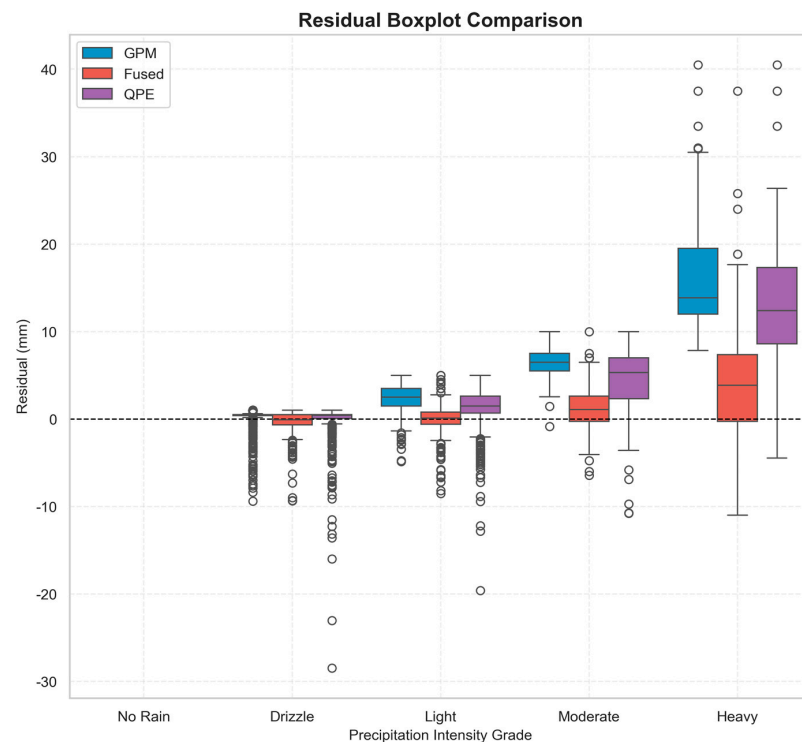


Figure 7. Residual distributions of precipitation estimates before and after fusion.

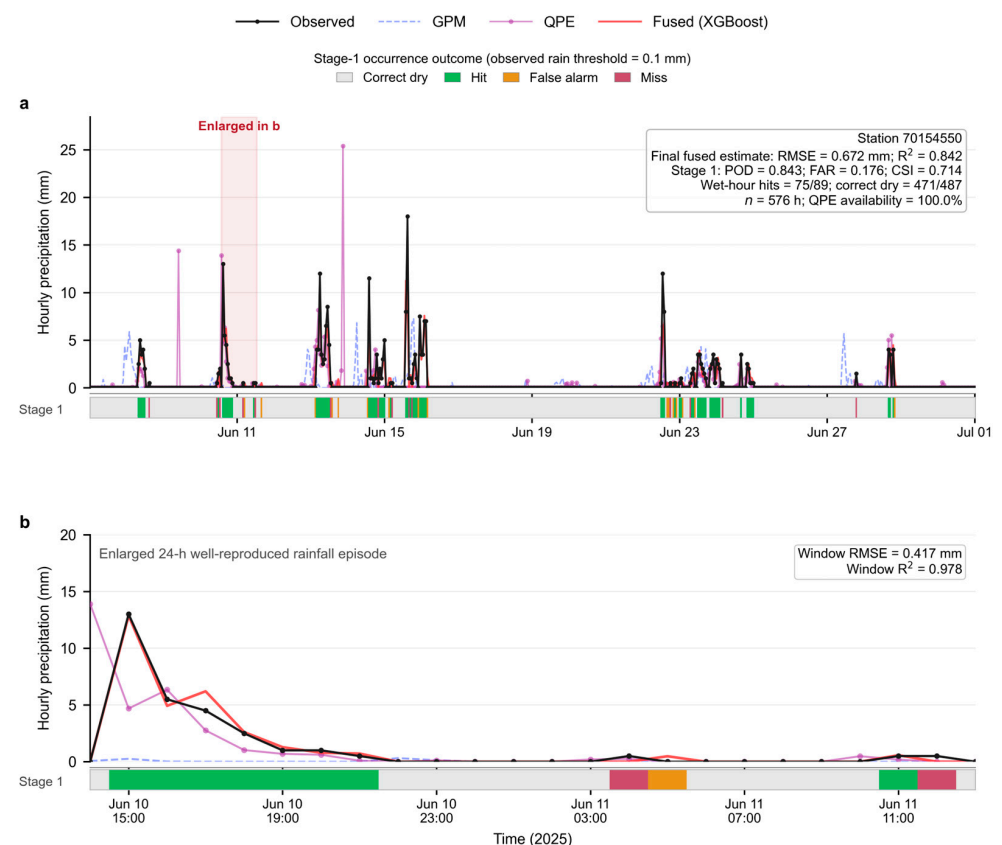


Figure 8. Event-scale precipitation comparison at held-out Station 70154550. (a) Complete 576 h evaluation record; the shaded region marks the 24 h interval enlarged in (b). (b) Enlarged view of the rainfall episode from 14:00 on 10 June to 13:00 on 11 June 2025. The strips show Stage-1 occurrence outcomes using observed precipitation ≥ 0.1 mm as the rain threshold. Metrics in (a) and (b) refer to the complete record and enlarged interval, respectively. The complete five-station comparison is shown in Figure A1; aggregate performance is based on all 15 held-out stations.

Panel (b) enlarges a 24 h rainfall episode from 14:00 on 10 June to 13:00 on 11 June 2025. Within this interval, the fused product achieved an RMSE of 0.417 mm and an R^2 of 0.978. The 13.0 mm observed peak was estimated as 12.86 mm by the fused product, compared with 0.25 mm for GPM and 4.69 mm for radar QPE. The fused estimates closely followed the observed occurrence and intensity during this interval, whereas the complete record retained visible misses, false alarms, and residual intensity errors.

Additional time-series diagnostics for five illustrative held-out stations are provided in Appendix B and Figure A1. Figure 8 provides a process-level illustration at one station, whereas aggregate model performance is based on all 15 held-out test stations.

At the spatial scale, GPM displayed coarse block-like patterns and did not resolve several local rainfall maxima observed near the southern mountainous area. Radar QPE provided fine spatial textures but showed missing or underestimated values in areas affected by terrain shielding. By combining radar detail, satellite background, and terrain-interaction features, the fusion product produced spatial patterns that were more consistent with the observed station-level rainfall distribution and local topographic gradients. The spatial patterns of station-mean precipitation and radar QPE availability are further examined in Figure 9.

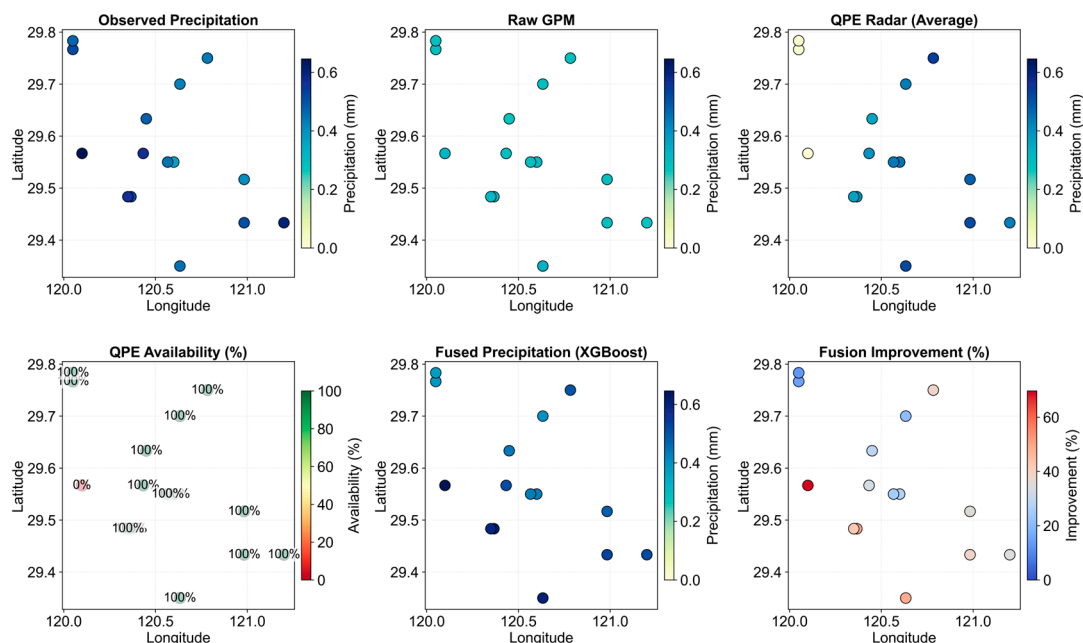


Figure 9. Spatial analysis of station-mean precipitation and radar QPE availability.

4.6. Feature Importance and Physical Interpretation

Figure 10 presents the top 20 feature-importance values of the fusion model. Radar QPE-related variables were prominent among the top-ranked predictors, indicating that radar information was heavily used by the fitted model in the present case study. Satellite-derived variables supplied large-scale precipitation-background information, whereas lagged and accumulated variables represented short-term rainfall persistence and storm evolution. Terrain-related predictors provided terrain-dependent information for precipitation correction, and neighborhood statistics represented local spatial coherence.

Feature-importance scores describe how predictors were used by the fitted model, but they do not establish causal effects or isolate the independent contribution of each predictor group. The results therefore indicate complementary information across predictor groups rather than direct attribution of physical precipitation processes. A group-wise ablation analysis would be required to quantify their independent contributions.

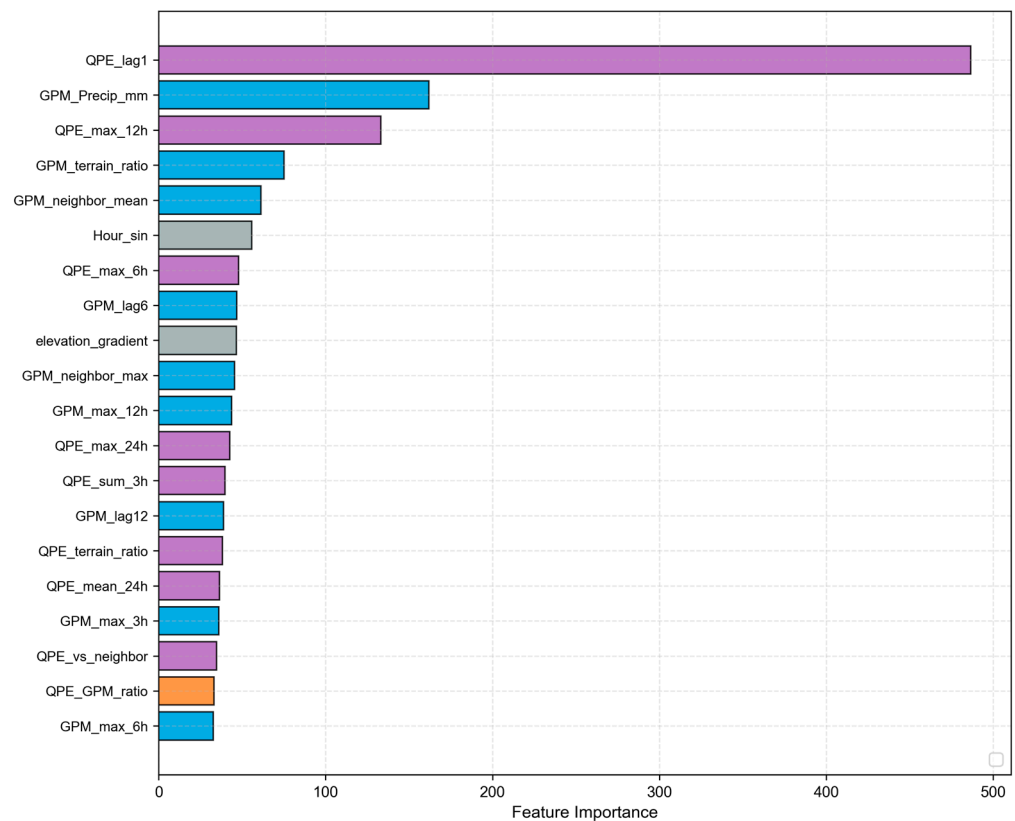


Figure 10. Top-20 feature importance values of the fusion model.

5. Discussion

5.1. Why the Two-Stage Framework Improves Fusion Performance

Despite the overall F1 score of 0.823, the first-stage LightGBM classifier produced 259 missed detections (FN) out of 1332 actual rainy samples (19.4% miss rate). Among the 259 missed detections, 175 (67.6%) occurred in the 0.5–1.0 mm class, corresponding to a miss rate of 42.0% for this intensity bin, while miss rates for events >1.0 mm ranged from 5.6% to 10.6%. Notably, the test set contains no observed events between 0.1 and 0.5 mm, so the previous inference that “all FN ≥ 0.5 mm” indicates perfect trace-event detection is a data-distribution artifact rather than a valid conclusion.

Analysis of predictor availability reveals that 50.6% of the FN samples have GPM precipitation equal to zero (satellite retrieval completely missed the event), and 35.5% have QPE either equal to zero or unavailable. In 18.1% of FN cases, both GPM and QPE simultaneously registered zero or missing values. This indicates that the dominant cause of missed detections is not an overly conservative probability cutoff, but rather the failure of all input precipitation fields (satellite and radar) to register the event. When both primary predictors provide no rainfall signal, the LightGBM classifier—trained to avoid false positives—correctly assigns a low probability to the rainy class.

The rain gauges can capture highly localized convective cells that are not fully resolved by coarse-resolution satellite pixels or radar observations, highlighting the value of integrating complementary data sources to improve the representation of localized rainfall extremes. Among the missed cases, 15 events exceeded 5 mm, and 5 exceeded 10 mm, including one event reaching 37.5 mm—a flash-flood-relevant intensity. These heavy FN events typically occurred at stations where the 61-dimensional feature vector did not produce a sufficiently discriminative probability because the concurrent GPM and radar signals were anomalously low. Users should be aware that the current classification probability

cutoff favors operational reliability (low FAR) over extreme-event capture; a lower cutoff could be adopted for flash-flood early warning at the cost of increased false alarms.

The classification probability cutoff—optimized to approximately 0.42 via F1-score on the validation folds—directly controls the trade-off between detection rate and false alarm rate. Lowering this cutoff from 0.42 to 0.30 would increase recall from 0.806 to approximately 0.85 but raise FAR from 0.158 to 0.23, while raising it to 0.60 would reduce FAR to 0.12 but further increase missed heavy events.

The training dataset contains only ~1020 samples ≥ 5 mm and ~345 samples ≥ 10 mm out of ~33,000 total records (approximately 3.1% and 1.0%, respectively). This extreme class imbalance inherently limits the regression model's exposure to intense convective events [42]. Although the XGBoost residual approach improves heavy-rainfall estimation substantially (reducing >10 mm bias from -16.86 mm to -5.19 mm), the residual underestimation of approximately 5 mm for extreme events (Section 4.2) reflects the limited sample size in the upper tail. This residual bias is primarily associated with the limited representation of extreme events in the current training dataset and could be further reduced by extending the training period to multiple years or adopting targeted sampling strategies for the upper intensity tail.

The QPE_available flag provides a natural setting to evaluate model behavior when radar information is absent (576 samples, 6.7% of the test set). In radar-blind conditions, the fused RMSE increased from 1.110 mm to 1.989 mm (+79%), and the mean bias shifted from near-zero (-0.006 mm) to -0.013 mm. This degradation indicates that radar QPE provided valuable spatial detail for localizing convective rainfall in this study [18,19]. However, the model still substantially outperformed GPM alone (RMSE = 3.011 mm in radar-blind zones), indicating that temporal-lag and satellite background features partially compensated for missing radar information. The framework therefore retained partial skill under radar-blind conditions but could not fully substitute for the missing structural information. This dependence on radar availability means that model performance may be lower in regions where radar coverage is systematically incomplete or where radar errors differ from those observed during the 2025 Shaoxing flood-season case. These findings are consistent with recent studies showing that machine-learning models can improve precipitation estimation by exploiting spatio-temporal information and radar-derived predictors in complex terrain [33,35]. However, unlike approaches that focus primarily on radar-based QPE or satellite–gauge merging, the present framework explicitly separates precipitation occurrence detection from conditional intensity correction. This design is particularly relevant for hourly rainfall data because it reduces the influence of zero-inflated samples on intensity estimation while retaining information from multiple precipitation sources.

The 72 gauges are unevenly distributed, with dense coverage in the northern plains (approximately 1 station per 50 km²) and sparse coverage in the southern mountainous areas (<1 station per 200 km²). The 15 held-out test stations exhibited a wide range of fusion RMSE values (0.478–2.137 mm), and stations with lower QPE availability tended to have higher RMSE. This spatial heterogeneity indicates that model skill was not uniform across the study area; errors in ungauged mountainous locations may be larger than the held-out test-set average. Additional gauges in data-sparse southern catchments would enable a more stringent evaluation of spatial generalization and may reveal failure modes not represented by the present station network.

5.2. Role of Terrain-Aware Feature Engineering

The results highlight the importance of explicitly encoding terrain and spatial-neighborhood information. In complex terrain, precipitation errors are not purely statistical

but are associated with physical mechanisms such as orographic lifting, radar beam blockage and spatially localized convective development. By introducing elevation, slope, terrain ratios and slope-interaction terms, the machine-learning model gained access to physically meaningful descriptors that helped it learn spatially variable correction functions.

Temporal-lag and accumulation features were designed to represent rainfall persistence and recent storm evolution, while terrain and neighborhood descriptors provided spatial context for precipitation correction. This design brings together three complementary lines of recent research: terrain-informed satellite downscaling [32], radar-centered QPE and radar–gauge integration [33,34], and spatiotemporal tree-based satellite–gauge merging [35]. Unlike approaches developed primarily for a single data source, scale, or estimation task, the present framework integrates satellite precipitation, radar QPE, gauge observations, terrain descriptors, temporal-memory variables, and neighborhood statistics at 1 km/1 h, while separating precipitation-occurrence detection from conditional intensity correction. The framework therefore complements, rather than replaces, these specialized approaches.

5.3. Implications for Rain Gauge Network Optimization

The spatially heterogeneous error patterns revealed by the fusion framework may provide useful clues for future rain-gauge network optimization in complex terrain. The analysis in Section 5.1 shows that station-level fusion RMSE varies by a factor of approximately four (0.478–2.137 mm) across the 15 test stations, with larger errors concentrated in areas of lower QPE availability and higher topographic complexity. This indicates that the current network, which is densest in the northern plains (~1 station per 50 km²) and sparsest in the southern mountains (<1 station per 200 km²), does not provide uniform validation capability across the region.

The fusion product may serve as a preliminary indicator of spatially varying precipitation uncertainty and help identify candidate areas for new gauge deployment. Specifically, grid cells where the model shows large residual errors and relies heavily on GPM background fields in the absence of radar coverage (Section 5.1) may represent locations where an additional gauge could provide relatively high marginal information gain. In radar-blind zones such as the western part of the study area (8 stations with 0% QPE availability), additional gauges would be particularly valuable for constraining satellite retrieval bias and the fusion model's extrapolation error.

Beyond Shaoxing, the same sequence of precipitation fusion, spatial-error mapping, and identification of data-sparse subregions could be evaluated as a candidate workflow for rain-gauge network assessment in other complex-terrain basins. Its transferability was not tested in the present study and requires independent multi-year and cross-basin evaluation.

5.4. Limitations and Future Work

The temporal coverage and validation design define the principal boundary of the present evidence. The model was developed and evaluated using a single flood season in 2025. Although this period included several intense rainfall episodes, it cannot represent the full range of interannual variability, storm types, or precipitation conditions outside the flood season. Moreover, the development and held-out stations sampled the same regional event sequence. The reported results therefore characterize station-level spatial interpolation within the observed Shaoxing flood-season period rather than transfer to independent seasons, storm events, or external basins. Multi-year, event-independent, and cross-basin evaluations are needed to establish broader applicability.

Model performance also depends on the coverage and quality of radar QPE. Beam blockage, ground clutter, incomplete retrievals, and differences among radar-network

configurations may reduce the value of radar-derived predictors. Satellite background fields, temporal-memory variables, and terrain descriptors may provide partial information when radar data are unavailable, but they cannot be assumed to replace the missing fine-scale radar structure. The results obtained in radar-blind or intermittently covered areas should therefore be interpreted cautiously. Evaluation across independent events and regions with different radar-network configurations is needed to determine how strongly the framework depends on radar coverage and data quality.

Several additional limitations remain. The current feature matrix relies primarily on precipitation products and static terrain descriptors. Atmospheric predictors such as wind profiles, water-vapor flux, convective available potential energy, and cloud-microphysical variables may improve the representation of precipitation conditions. In addition, the framework corrects observed or near-real-time precipitation fields but does not directly perform nowcasting. Future work could combine the proposed fusion logic with sequence-based models to support integrated monitoring and nowcasting applications [53,54].

Recent reviews have identified uncertainty quantification, additional remote-sensing variables, hybrid physical-machine-learning models, and transfer learning as important directions for improving precipitation estimation in complex and data-scarce regions [55]. Integrating these approaches in future work could further enhance the framework's physical interpretability, uncertainty characterization, and cross-region transferability.

6. Conclusions

This study proposes a two-stage machine learning fusion framework for 1 km/1 h precipitation estimation over Shaoxing, China, integrating 72 rain gauges, GPM IMERG V07 satellite data, S-band radar QPE, and DEM-derived terrain features via a LightGBM occurrence classifier and a Bayesian-optimized XGBoost residual regressor.

The first-stage classifier effectively filtered zero-inflated noise and false weak rainfall. After model selection using station-blocked fivefold cross-validation within the 57-station development set, the occurrence classifier was evaluated on the fixed 15-station held-out spatial test set, where it achieved an accuracy of 0.947, a positive-class F1 score of 0.823, precision of 0.842, and recall of 0.806.

On the same held-out test set, the fused precipitation product substantially improved quantitative accuracy. RMSE decreased from 2.342 mm for GPM to 1.189 mm for the XGBoost fusion model, corresponding to a 49.22% reduction, while MAE decreased from 0.712 mm to 0.262 mm and R^2 increased to 0.685.

The two-stage framework improved heavy-rainfall representation and reduced false weak precipitation in the fixed held-out test set. Feature-importance analysis indicated that radar QPE, satellite background precipitation, temporal-memory variables, terrain-related predictors, and neighborhood statistics provided complementary information within the fitted model; these results should not be interpreted as direct attribution of physical precipitation processes or as a substitute for group-wise ablation analysis.

Overall, the proposed approach provides a practical case-study workflow for 1 km/1 h precipitation fusion over Shaoxing. The fused product and its spatial error diagnostics may support the evaluation of rain-gauge network gaps in data-sparse mountainous areas. The present findings should nevertheless be interpreted as evidence from the 2025 Shaoxing flood-season case, and broader applicability across independent seasons, storm events, and different hydroclimatic or radar-network settings requires further validation.

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L.C.; visualization, L.C. and J.Q.; supervision, S.W.; project administration, H.W.; funding acquisition, S.W. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Not applicable.

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Data Availability Statement: The GPM IMERG Late Run V07 precipitation data used in this study are publicly available from NASA GES DISC/PPS. The SRTM DEM data are publicly available from NASA Earthdata/LP DAAC. The rain-gauge observations and operational radar QPE data used in this study were provided by the relevant hydrometeorological agencies and are not publicly available due to data-use restrictions. The derived datasets are available from the corresponding author upon reasonable request, subject to approval by the original data providers.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BO	Bayesian optimization
CART	Classification and regression tree
CC	Correlation coefficient
CSI	Critical success index
DEM	Digital elevation model
FAR	False alarm ratio
FN	False negative
FP	False positive
GPM	Global Precipitation Measurement
IMERG	Integrated Multi-satellitE Retrievals for GPM
LightGBM	Light Gradient Boosting Machine
MAE	Mean absolute error
NWP	Numerical weather prediction
PBIAS	Percent bias
POD	Probability of detection
QPE	Quantitative precipitation estimation
R ²	Coefficient of determination
RMSE	Root mean square error
RMS	Root mean square
SRTM	Shuttle Radar Topography Mission
TP	True positive
XGBoost	eXtreme Gradient Boosting

Appendix A

The complete predictor list used in the two-stage precipitation-fusion model is provided in Table A1.

Table A1. Complete list of the 61 predictors used in the two-stage precipitation-fusion model. Static terrain variables were derived from the SRTM DEM; dynamic precipitation variables were derived from GPM IMERG and radar QPE after temporal aggregation and spatial matching.

No.	Predictor	Symbol/Unit	Window	Definition/Calculation
Static geography and terrain forcing				
1	Lon	lon_i ; degree	Static	Longitude of station i or the matched 1 km grid cell.
2	Lat	lat_i ; degree	Static	Latitude of station i or the matched 1 km grid cell.
3	Elevation	z_i ; m	Static	Elevation derived from the SRTM DEM and aggregated to the 1 km grid.
4	Slope	s_i ; degree	Static	Local terrain slope derived from the SRTM DEM and aggregated to the 1 km grid.
Cyclic and radiative timing				
5	Hour_sin	$\sin(2\pi \text{ hour}_t/24)$; dimensionless	Current hour	Sine transformation of hour of day to preserve cyclic continuity.
6	Hour_cos	$\cos(2\pi \text{ hour}_t/24)$; dimensionless	Current hour	Cosine transformation of hour of day to preserve cyclic continuity.
7	Month_sin	$\sin(2\pi \text{ month}_t/12)$; dimensionless	Current month	Sine transformation of month to represent seasonal phase.
8	Month_cos	$\cos(2\pi \text{ month}_t/12)$; dimensionless	Current month	Cosine transformation of month to represent seasonal phase.
9	Is_Night	$I_{\text{night}}(t)$; 0/1	Current hour	Binary indicator of nighttime conditions used to represent diurnal radiative contrast.
Multi-source initial state				
10	GPM_Precip	$P_{i,t}^{\text{GPM}}$; mm h ⁻¹	Current hour	Hourly GPM IMERG Late Run precipitation matched to the 1 km grid.
11	QPE_Precip	$P_{i,t}^{\text{QPE}}$; mm h ⁻¹	Current hour	Hourly radar QPE precipitation matched to the 1 km grid; unavailable values were filled with 0 mm before model input.
12	QPE_available	$A_{i,t}^{\text{QPE}}$; 0/1	Current hour	Radar-availability flag: 1 for valid radar retrieval including true zero precipitation, and 0 for no coverage, failed retrieval, or insufficient hourly completeness.
Temporal memory				
13	GPM_lag_1h	$P_{i,t-1}^{\text{GPM}}$; mm h ⁻¹	$t - 1$ h	GPM precipitation at station/grid cell i lagged by 1 h.
14	GPM_lag_3h	$P_{i,t-3}^{\text{GPM}}$; mm h ⁻¹	$t - 3$ h	GPM precipitation at station/grid cell i lagged by 3 h.
15	GPM_lag_6h	$P_{i,t-6}^{\text{GPM}}$; mm h ⁻¹	$t - 6$ h	GPM precipitation at station/grid cell i lagged by 6 h.
16	GPM_lag_12h	$P_{i,t-12}^{\text{GPM}}$; mm h ⁻¹	$t - 12$ h	GPM precipitation at station/grid cell i lagged by 12 h.
17	QPE_lag_1h	$P_{i,t-1}^{\text{QPE}}$; mm h ⁻¹	$t - 1$ h	QPE precipitation at station/grid cell i lagged by 1 h.
18	QPE_lag_3h	$P_{i,t-3}^{\text{QPE}}$; mm h ⁻¹	$t - 3$ h	QPE precipitation at station/grid cell i lagged by 3 h.
19	QPE_lag_6h	$P_{i,t-6}^{\text{QPE}}$; mm h ⁻¹	$t - 6$ h	QPE precipitation at station/grid cell i lagged by 6 h.
20	QPE_lag_12h	$P_{i,t-12}^{\text{QPE}}$; mm h ⁻¹	$t - 12$ h	QPE precipitation at station/grid cell i lagged by 12 h.

Table A1. Cont.

No.	Predictor	Symbol/Unit	Window	Definition/Calculation
Accumulation and extremes				
21	GPM_sum_3h	$A_{i,t}^{(3,GPM)}; \text{mm}$	3 h ending at t	Moving-window accumulation of GPM precipitation over the previous 3 h ending at hour t.
22	GPM_sum_6h	$A_{i,t}^{(6,GPM)}; \text{mm}$	6 h ending at t	Moving-window accumulation of GPM precipitation over the previous 6 h ending at hour t.
23	GPM_sum_12h	$A_{i,t}^{(12,GPM)}; \text{mm}$	12 h ending at t	Moving-window accumulation of GPM precipitation over the previous 12 h ending at hour t.
24	GPM_sum_24h	$A_{i,t}^{(24,GPM)}; \text{mm}$	24 h ending at t	Moving-window accumulation of GPM precipitation over the previous 24 h ending at hour t.
25	QPE_sum_3h	$A_{i,t}^{(3,QPE)}; \text{mm}$	3 h ending at t	Moving-window accumulation of QPE precipitation over the previous 3 h ending at hour t.
26	QPE_sum_6h	$A_{i,t}^{(6,QPE)}; \text{mm}$	6 h ending at t	Moving-window accumulation of QPE precipitation over the previous 6 h ending at hour t.
27	QPE_sum_12h	$A_{i,t}^{(12,QPE)}; \text{mm}$	12 h ending at t	Moving-window accumulation of QPE precipitation over the previous 12 h ending at hour t.
28	QPE_sum_24h	$A_{i,t}^{(24,QPE)}; \text{mm}$	24 h ending at t	Moving-window accumulation of QPE precipitation over the previous 24 h ending at hour t.
29	GPM_max_3h	$M_{i,t}^{(3,GPM)}; \text{mm h}^{-1}$	3 h ending at t	Maximum hourly GPM precipitation within the previous 3 h window ending at hour t.
30	GPM_max_6h	$M_{i,t}^{(6,GPM)}; \text{mm h}^{-1}$	6 h ending at t	Maximum hourly GPM precipitation within the previous 6 h window ending at hour t.
31	GPM_max_12h	$M_{i,t}^{(12,GPM)}; \text{mm h}^{-1}$	12 h ending at t	Maximum hourly GPM precipitation within the previous 12 h window ending at hour t.
32	GPM_max_24h	$M_{i,t}^{(24,GPM)}; \text{mm h}^{-1}$	24 h ending at t	Maximum hourly GPM precipitation within the previous 24 h window ending at hour t.
33	QPE_max_3h	$M_{i,t}^{(3,QPE)}; \text{mm h}^{-1}$	3 h ending at t	Maximum hourly QPE precipitation within the previous 3 h window ending at hour t.
34	QPE_max_6h	$M_{i,t}^{(6,QPE)}; \text{mm h}^{-1}$	6 h ending at t	Maximum hourly QPE precipitation within the previous 6 h window ending at hour t.
35	QPE_max_12h	$M_{i,t}^{(12,QPE)}; \text{mm h}^{-1}$	12 h ending at t	Maximum hourly QPE precipitation within the previous 12 h window ending at hour t.
36	QPE_max_24h	$M_{i,t}^{(24,QPE)}; \text{mm h}^{-1}$	24 h ending at t	Maximum hourly QPE precipitation within the previous 24 h window ending at hour t.
Statistical variability and anomalies				
37	GPM_mean_24h	$\mu_{i,t}^{(24,GPM)}; \text{mm h}^{-1}$	24 h ending at t	Mean hourly GPM precipitation over the previous 24 h ending at hour t.
38	QPE_mean_24h	$\mu_{i,t}^{(24,QPE)}; \text{mm h}^{-1}$	24 h ending at t	Mean hourly QPE precipitation over the previous 24 h ending at hour t.
39	GPM_std_24h	$\sigma_{i,t}^{(24,GPM)}; \text{mm h}^{-1}$	24 h ending at t	Standard deviation of hourly GPM precipitation over the previous 24 h ending at hour t.

Table A1. Cont.

No.	Predictor	Symbol/Unit	Window	Definition/Calculation
40	QPE_std_24h	$\sigma_{i,t}^{(24,QPE)}; \text{mm h}^{-1}$	24 h ending at t	Standard deviation of hourly QPE precipitation over the previous 24 h ending at hour t.
41	GPM_deviation_24h	$P_{i,t}^{\text{GPM}} - \mu_{i,t}^{(24,\text{GPM})}; \text{mm h}^{-1}$	Current minus 24 h mean	Deviation of current GPM precipitation from its 24 h moving mean.
42	QPE_deviation_24h	$P_{i,t}^{\text{QPE}} - \mu_{i,t}^{(24,\text{QPE})}; \text{mm h}^{-1}$	Current minus 24 h mean	Deviation of current QPE precipitation from its 24 h moving mean.
Spatial-neighborhood dependence				
43	GPM_neighbor_mean	$\text{bar}(P)_{i,t}^{(5,\text{GPM})}; \text{mm h}^{-1}$	Current hour, K = 5	Mean precipitation of the five nearest neighboring 1 km grid cells for the source field.
44	GPM_neighbor_max	$M_{i,t}^{(5,\text{GPM})}; \text{mm h}^{-1}$	Current hour, K = 5	Maximum precipitation among the five nearest neighboring 1 km grid cells for the source field.
45	GPM_neighbor_std	$\sigma_{i,t}^{(5,\text{GPM})}; \text{mm h}^{-1}$	Current hour, K = 5	Standard deviation of precipitation among the five nearest neighboring 1 km grid cells for the source field.
46	QPE_neighbor_mean	$\text{bar}(P)_{i,t}^{(5,\text{QPE})}; \text{mm h}^{-1}$	Current hour, K = 5	Mean precipitation of the five nearest neighboring 1 km grid cells for the source field.
47	QPE_neighbor_max	$M_{i,t}^{(5,\text{QPE})}; \text{mm h}^{-1}$	Current hour, K = 5	Maximum precipitation among the five nearest neighboring 1 km grid cells for the source field.
48	QPE_neighbor_std	$\sigma_{i,t}^{(5,\text{QPE})}; \text{mm h}^{-1}$	Current hour, K = 5	Standard deviation of precipitation among the five nearest neighboring 1 km grid cells for the source field.
Terrain and multi-source interaction				
49	QPE_GPM_ratio	$P_{i,t}^{\text{QPE}} / (P_{i,t}^{\text{GPM}} + \epsilon); \text{dimensionless}$	Current hour	Ratio between radar QPE and GPM precipitation; epsilon prevents division by zero.
50	QPE_GPM_difference	$P_{i,t}^{\text{QPE}} - P_{i,t}^{\text{GPM}}; \text{mm h}^{-1}$	Current hour	Difference between radar QPE and GPM precipitation.
51	QPE_GPM_mean	$(P_{i,t}^{\text{QPE}} + P_{i,t}^{\text{GPM}})/2; \text{mm h}^{-1}$	Current hour	Arithmetic mean of the two precipitation-source estimates.
52	QPE_GPM_max	$\max(P_{i,t}^{\text{QPE}}, P_{i,t}^{\text{GPM}}); \text{mm h}^{-1}$	Current hour	Maximum of radar QPE and GPM precipitation.
53	GPM_terrain_ratio	$P_{i,t}^{\text{GPM}} \times z_i^*; \text{mm h}^{-1}$	Current hour + static terrain	GPM precipitation scaled by normalized elevation z_i^* to represent terrain-modulated satellite bias.
54	QPE_terrain_ratio	$P_{i,t}^{\text{QPE}} \times z_i^*; \text{mm h}^{-1}$	Current hour + static terrain	Radar QPE precipitation scaled by normalized elevation z_i^* to represent terrain-modulated radar error.
55	GPM_slope_interaction	$P_{i,t}^{\text{GPM}} \times s_i^*; \text{mm h}^{-1}$	Current hour + static terrain	Interaction between GPM precipitation and normalized slope s_i^* .
56	QPE_slope_interaction	$P_{i,t}^{\text{QPE}} \times s_i^*; \text{mm h}^{-1}$	Current hour + static terrain	Interaction between radar QPE precipitation and normalized slope s_i^* .
57	Elevation_gradient	$\text{grad}(z)_i; \text{m km}^{-1}$	Static neighborhood	Local elevation gradient calculated from the surrounding 1 km grid cells.

Table A1. Cont.

No.	Predictor	Symbol/Unit	Window	Definition/Calculation
58	GPM_elevation_gradient	$P_{i,t}^{\text{GPM}} \times \text{grad}(z)_i;$ mm h ⁻¹	Current hour + static neighborhood	Interaction between GPM precipitation and local elevation gradient.
59	QPE_elevation_gradient	$P_{i,t}^{\text{QPE}} \times \text{grad}(z)_i;$ mm h ⁻¹	Current hour + static neighborhood	Interaction between radar QPE precipitation and local elevation gradient.
60	GPM_local_anomaly	$P_{i,t}^{\text{GPM}} - \text{bar}(P)_{i,t}^{(5,\text{GPM})};$ mm h ⁻¹	Current hour, K = 5	Local anomaly of GPM precipitation relative to its five-neighbor mean.
61	QPE_local_anomaly	$P_{i,t}^{\text{QPE}} - \text{bar}(P)_{i,t}^{(5,\text{QPE})};$ mm h ⁻¹	Current hour, K = 5	Local anomaly of radar QPE precipitation relative to its five-neighbor mean.

Note: z_i^* and s_i^* denote normalized elevation and normalized slope, respectively; ε is a small positive constant used to avoid division by zero. K = 5 indicates the five nearest neighboring 1 km grid cells.

Appendix B

Figure A1 Station-level time-series comparisons compare the hourly precipitation series at five illustrative held-out stations. Across these stations, the root mean square error of the fused product ranged from 0.672 to 1.989 mm, with coefficients of determination ranging from 0.511 to 0.842. The corresponding GPM coefficients of determination were negative at all five stations, indicating that the satellite product reproduced the station-level hourly variability poorly during the evaluation period. The fused series more closely followed the timing and magnitude of most observed rainfall episodes, although several of the largest observed peaks remained underestimated. The Stage-1 occurrence strips additionally distinguish correct dry hours, hits, false alarms, and misses using an observed-rain threshold of 0.1 mm. Across the five stations, probability of detection ranged from 0.808 to 0.906, false alarm ratio from 0.103 to 0.200, and critical success index from 0.672 to 0.821. These panels provide station-level process diagnostics and do not replace the aggregate performance evaluation across all 15 held-out stations.

The contrast among stations also illustrates the influence of radar availability and local error heterogeneity. At Station 70117250, no concurrent radar QPE was available during the evaluation period. The fused product reduced RMSE from 3.011 mm for GPM to 1.989 mm, but it reproduced only approximately 20.08 mm of the maximum 51 mm observed peak. This result is consistent with partial compensation by the satellite background and other predictors when radar information is absent, but it does not imply that these inputs fully replace the missing radar structure or isolate the contribution of any individual feature group. At the other four stations, where QPE availability was 100%, fused-product RMSE ranged from 0.672 to 1.243 mm.

Among the five illustrative stations, Station 70154550 had the lowest RMSE and highest R^2 . Its detailed time-series comparison is presented in Figure 8, while aggregate conclusions remain based on the complete held-out test set.

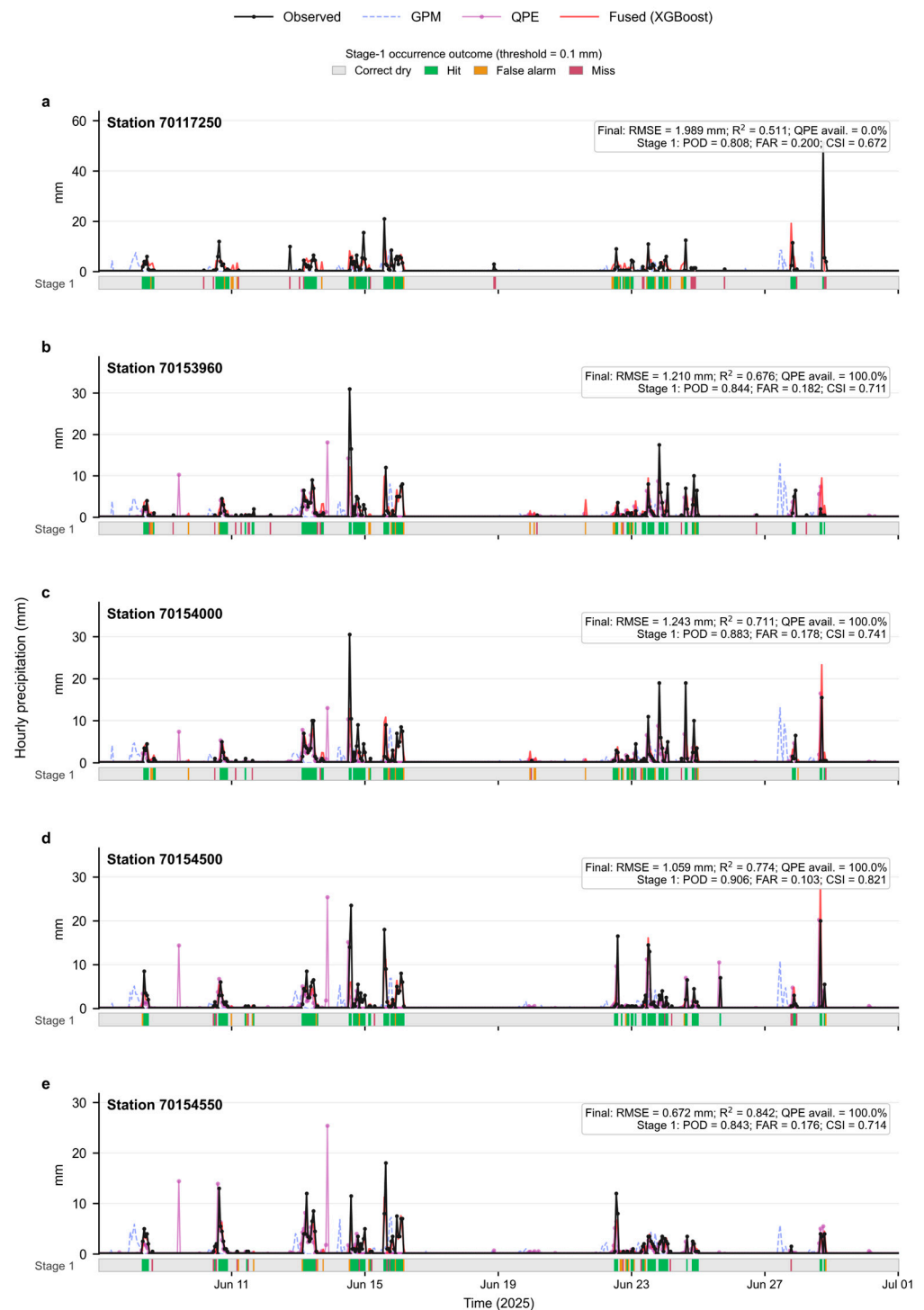


Figure A1. Hourly precipitation comparisons at five held-out stations. (a) Station 70117250; (b) Station 70153960; (c) Station 70154000; (d) Station 70154500; and (e) Station 70154550. Stage-1 occurrence outcomes are shown below each time series using an observed-rain threshold of 0.1 mm. Metrics are based on the complete 576 h record at each station. Panel-specific y-axis limits are used to preserve within-station detail, and radar QPE is omitted when unavailable.

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